

Title: Crop Yield Prediction Using Machine Learning

Project Report

Prashik Kadam

2026



Fundamentals of AI Using Agriculture Data Set

Certificate

This is to certify that Prashik Kadam from Pandit Dwarka Prasad Mishra Indian Institute Of Information Technology Design And Manufacturing Jabalpur has successfully completed the project work titled Crop Yield Prediction Using Machine Learning as part of the 1-credit course “Fundamentals of AI using Agriculture Data Set”, offered under ANNAM.AI – Centre of Excellence, Ministry of Education, Government of India, at IIT Ropar. This project constitutes a component of the 30 hours of learning prescribed for the course and reflects the student’s independent effort in problem selection, method design, and implementation.

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1. Introduction and Motivation

Agriculture plays a vital role in the economy and food production of many countries. Predicting crop yield accurately can help farmers, agricultural organizations, and policymakers make better decisions regarding crop planning and resource management.

This project focuses on predicting crop yield using Machine Learning techniques based on environmental and agricultural factors such as rainfall, pesticide usage, temperature, crop type, and geographical area.

The project demonstrates the application of Artificial Intelligence and data analytics in solving real-world agricultural prediction problems and promoting smart farming practices.

2. Problem Statement

The main objective of this project is to predict crop yield using multiple agricultural and environmental features.

The task involves building a regression-based machine learning model that can learn patterns from historical agricultural data and estimate crop productivity accurately.

The prediction model can help improve agricultural planning and support data-driven farming decisions.

3. Dataset Understanding

The dataset used in this project is the Crop Yield Prediction dataset.

It contains several attributes related to agriculture and environmental conditions such as:

- Area
- Crop Type
- Year
- Average Rainfall
- Pesticide Usage
- Average Temperature
- Crop Yield (Target Variable).

The dataset was explored using descriptive statistics, visualization techniques, and missing value analysis to understand relationships among different features.

4. Methodology

The following methodology was followed:

- Data collection and loading
- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature encoding
- Feature selection
- Splitting dataset into training and testing sets
- Training machine learning models
- Evaluating model performance using MAE, RMSE, and R^2 Score
- Comparing model performance
- Visualizing predictions and feature importance

5. Implementation Details

The implementation was carried out using Python in Google Colab/Jupyter Notebook.

Libraries used include:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Two machine learning models were trained:

- Linear Regression
- Random Forest Regressor

Various visualizations such as heatmaps, scatter plots, distribution plots, feature importance

graphs, and prediction comparison graphs were generated to analyze the dataset and model performance.

The categorical features such as Area and Crop Type were encoded before training the models.

6. Results and Discussions

The machine learning models successfully learned patterns from the agricultural dataset and generated crop yield predictions.

Among the models used, the Random Forest Regressor provided better prediction accuracy compared to Linear Regression because it was able to capture nonlinear relationships between environmental factors and crop yield.

The performance of the models was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Visual analysis showed that rainfall, temperature, and pesticide usage significantly influenced crop productivity.

Challenges faced during the project:

- Handling categorical data encoding
- Data preprocessing and cleaning
- Understanding feature correlations

Skills learned:

- Data preprocessing
- Exploratory Data Analysis
- Machine learning workflow
- Regression modelling
- Agricultural data analysis
- Data visualization

7. Conclusion

This project demonstrated the use of supervised machine learning techniques for crop yield prediction.

The Random Forest Regressor model provided accurate predictions and helped in understanding how environmental and agricultural factors affect crop production.

The project improved practical knowledge of Artificial Intelligence, machine learning, data analytics, and smart agriculture systems.

The developed prediction system can support future smart farming solutions and assist farmers in improving agricultural productivity.

8. References

- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib Documentation
- Kaggle Crop Yield Prediction Dataset
- Google colab

Appendix

The appendix includes:

- Sample code snippets
- Visualization outputs
- Heatmaps
- Prediction graphs
- Feature importance graphs
- Actual vs Predicted plots

Thank You